# WK1-PM1 – WEEK 1 PROJECT MODULE 1

## LAB SETUP: VIRTUALBOX AND KALI LINUX

| | |
|---|---|
| **Project:** | Cybersecurity Testing Lab Environment Setup |
| **Task ID:** | WK1-PM1 |
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| **Course:** | Cybersecurity / Ethical Hacking |
| **Date of Completion:** | 29-10-2026 |

## 1. OBJECTIVE

The objective of this task is to set up a cybersecurity testing lab environment using Oracle VirtualBox and Kali Linux. The lab is designed for authorized cybersecurity learning, penetration-testing practice, and security testing in a controlled virtual environment.

The lab configuration includes a custom NAT Network, a static IP address for Kali Linux, internet connectivity, shared clipboard, drag-and-drop functionality, a shared Downloads folder, and a virtual machine snapshot.

## 2. SOFTWARE AND SYSTEM REQUIREMENTS

The following software and resources were used or required for the lab setup:

| Component | Details |
|---|---|
| **Host Operating System** | Microsoft Windows |
| **Virtualization Software** | Oracle VirtualBox |
| **Attacker Machine** | Kali Linux |
| **Archive Utility** | 7-Zip |
| **Network Type** | NAT Network |
| **Network Subnet** | 10.0.0.0/24 |
| **Kali Linux IP Address** | 10.0.0.2/24 |
| **Default Gateway** | 10.0.0.1 |
| **Shared Host Folder** | Windows Downloads folder |

**Screenshot 1:** *Insert a screenshot of VirtualBox Manager showing the Kali Linux virtual machine.*

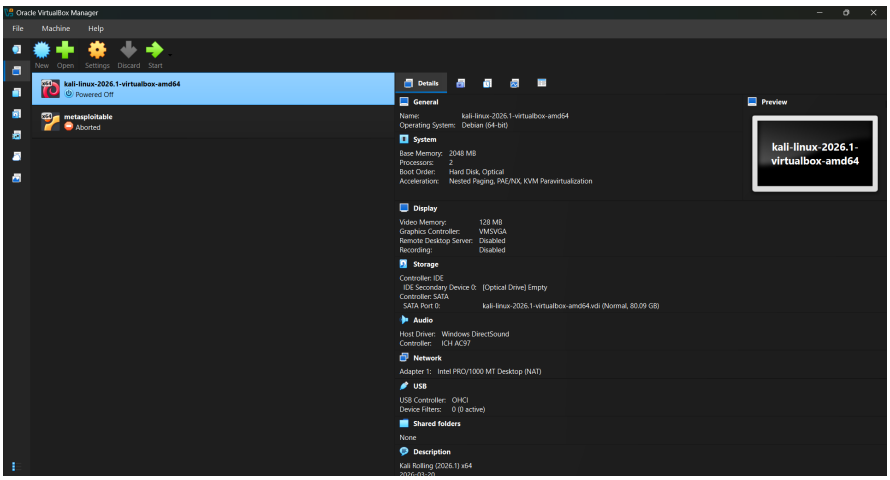
## 3. INSTALLATION OF VIRTUALBOX

Oracle VirtualBox was used as the virtualization platform for the cybersecurity lab.

### Procedure

1. Downloaded Oracle VirtualBox from the official website.
2. Installed VirtualBox on the Windows host computer.
3. Opened VirtualBox Manager and verified that the application was running.
4. Prepared the environment for importing the Kali Linux virtual machine.

**Official website:** https://www.virtualbox.org/wiki/Downloads




**Screenshot 3:** *Kali Linux virtual machine in VirtualBox Manager.*

## 4. DOWNLOAD AND IMPORT OF KALI LINUX

Kali Linux was selected as the attacker machine for the cybersecurity testing lab.

### Procedure

1. Accessed the official Kali Linux website.
2. Downloaded the prebuilt VirtualBox virtual machine image.
3. Extracted the downloaded archive using 7-Zip, if required.
4. Added the extracted virtual machine to VirtualBox Manager.
5. Configured the virtual machine's memory, processor, and network settings.
6. Started Kali Linux and verified that the operating system booted successfully.

**Official website:** https://www.kali.org/get-kali/#kali-virtual-machines




**Screenshot 4:** *Running Kali Linux desktop.*

## 5. NAT NETWORK CONFIGURATION

A custom NAT Network was configured to provide network connectivity to the virtual machines in the lab.

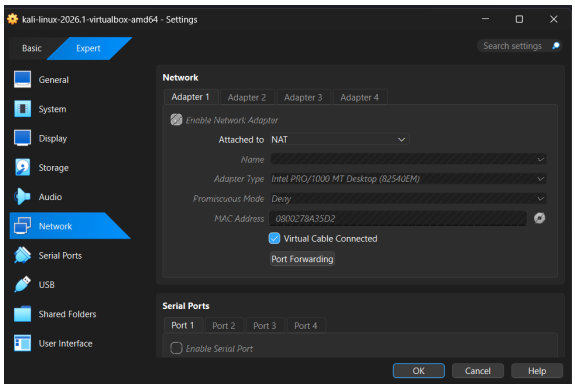
### Network Configuration

| Parameter | Configuration |
|---|---|
| **Network Name** | CyberLab-NAT |
| **Network Address** | 10.0.0.0 |
| **Subnet Prefix** | /24 |
| **Subnet Mask** | 255.255.255.0 |
| **Kali Linux IP Address** | 10.0.0.2 |
| **Default Gateway** | 10.0.0.1 |
| **Network Type** | NAT Network |

### Procedure

1. Opened VirtualBox Manager.
2. Navigated to the VirtualBox network management settings.
3. Created a custom NAT Network named CyberLab-NAT.
4. Configured the network prefix as 10.0.0.0/24.
5. Enabled the NAT Network.
6. Attached Kali Linux to the custom NAT Network through the virtual machine's network settings.

**Screenshot 5:** *Insert a screenshot of the NAT Network settings showing the network name and IPv4 prefix.*




**Screenshot 6:** *Kali Linux virtual machine settings showing Adapter 1 network attachment.*

## 6. KALI LINUX IP ADDRESS CONFIGURATION

The Kali Linux virtual machine was configured to use the static IPv4 address specified in the assignment.

### Configuration Details

- IP address: 10.0.0.2
- Prefix length: /24
- Subnet mask: 255.255.255.0
- Default gateway: 10.0.0.1
- DNS servers: 10.0.0.1 and 8.8.8.8

### Configuration Procedure

The active NetworkManager connection was identified using:

```
nmcli connection show
```

The connection was configured with the static IP address using:

```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.0.0.2/24 \
  ipv4.gateway 10.0.0.1 \
  ipv4.dns "10.0.0.1 8.8.8.8"
```

The connection was then activated:

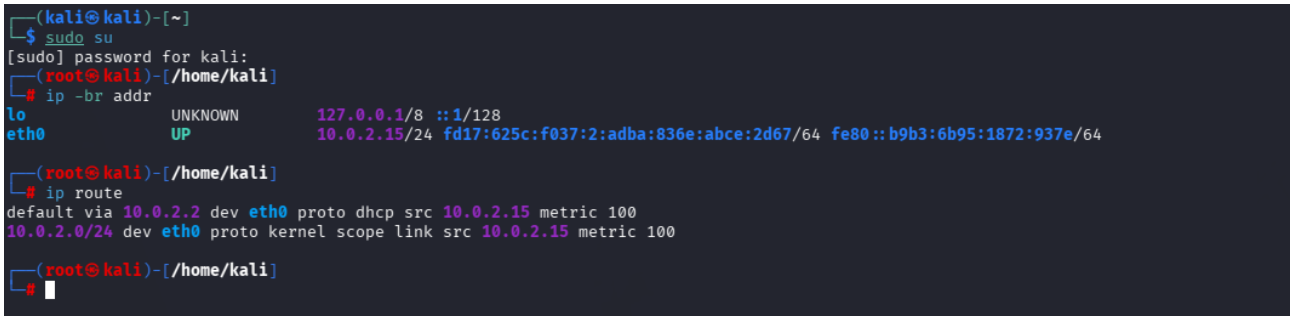
```
sudo nmcli connection up "Wired connection 1"
```

If the connection has a different name, the command must use the actual name shown by NetworkManager.

### Verification

The following commands were used to inspect the network configuration:

```
ip -br addr
ip route
```




**Screenshot 7:** *Kali Linux IP address and network interface, with the routing table.*

**Screenshot 8:** *Insert a screenshot showing the default gateway in the routing table.*

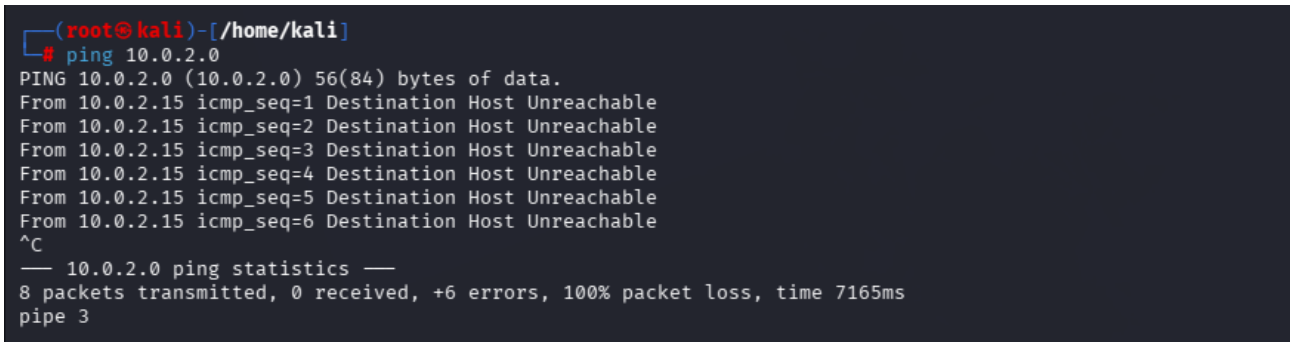
## 7. INTERNET CONNECTIVITY VERIFICATION

Internet connectivity was tested from the Kali Linux virtual machine.

### Test 1: Gateway Connectivity

```
ping -c 4 10.0.0.1
```

**Result:**




### Test 2: External IP Connectivity

```
ping -c 4 1.1.1.1
```

**Result:**

```
┌──(root㉿kali)-[/home/kali]
└─# ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=255 time=79.3 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=255 time=73.2 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=255 time=74.6 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=255 time=67.5 ms

── 1.1.1.1 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 67.532/73.642/79.272/4.187 ms
```

### Test 3: DNS Resolution

```
getent hosts kali.org
```

**Result:**

```
┌──(root㉿kali)-[/home/kali]
└─# getent hosts kali.org
2606:4700:83b0:747e:afe:91c:cfc8:4636 kali.org
```

### Test 4: HTTPS Connectivity

```
curl -I --max-time 15 https://www.kali.org/
```

```
┌──(root㉿kali)-[/home/kali]
└─# curl -I --max-time 15 https://www.kali.org/
HTTP/2 200
date: Fri, 30 Oct 2026 17:43:16 GMT
content-type: text/html; charset=utf-8
cache-control: max-age=600
x-frame-options: SAMEORIGIN
expires: Fri, 30 Oct 2026 17:53:16 UTC
last-modified: Mon, 26 Oct 2026 08:04:00 GMT
permissions-policy: interest-cohort=()
vary: Origin
vary: Accept-Encoding
x-request-id: 01M66YH9782MTMAR10VR950AJM
x-xss-protection: 1; mode=block
cf-cache-status: DYNAMIC
strict-transport-security: max-age=2592000
expect-ct: max-age=86400, enforce
referrer-policy: same-origin
x-content-type-options: nosniff
server: cloudflare
cf-ray: a52c38a3e8427f41-MAA
```

**Screenshot 9:** *HTTPS connectivity test result.*

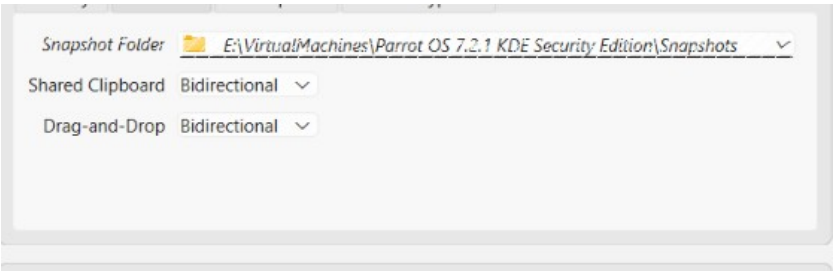
**Conclusion:** Internet connectivity was [verified / not yet verified] based on the test results recorded above.

## 8. SHARED CLIPBOARD AND DRAG-AND-DROP CONFIGURATION

Shared clipboard and drag-and-drop were configured to support data exchange between the Windows host and the Kali Linux virtual machine.

### Procedure

1. Opened the Kali Linux virtual machine settings in VirtualBox.
2. Navigated to General > Advanced.
3. Set Shared Clipboard to Bidirectional.
4. Set Drag'n'Drop to Bidirectional.
5. Saved the settings.
6. Tested text transfer between the host and guest operating systems.




**Screenshot 10:** *Shared clipboard and drag-and-drop settings.*

## 9. SHARED DOWNLOADS FOLDER CONFIGURATION

The Windows Downloads folder was configured as a shared folder for Kali Linux.

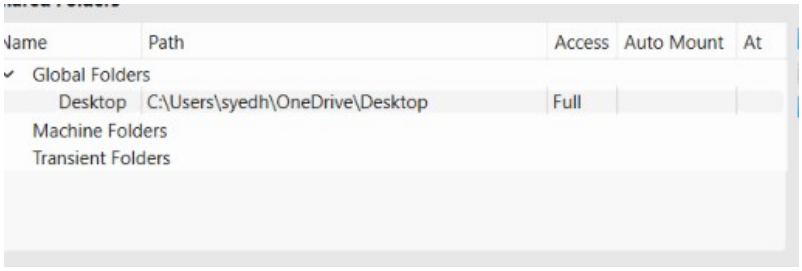
### Procedure

1. Opened Kali Linux virtual machine settings.
2. Selected Shared Folders.
3. Added the Windows Downloads folder.
4. Set the shared folder name to downloads.
5. Enabled Auto-mount and Make Permanent.
6. Saved the settings.
7. Checked the shared-folder mount inside Kali Linux.

### Verification Command

```
ls -la /media/sf_downloads
```

**Observed Result:** [Record whether the shared folder was mounted and accessible.]




**Screenshot 11:** *VirtualBox shared-folder configuration.*

## 10. VIRTUAL MACHINE SNAPSHOT

A snapshot was created to preserve the configured Kali Linux lab environment.

### Procedure

1. Opened VirtualBox Manager.
2. Selected the Kali Linux virtual machine.
3. Opened the Snapshots view.
4. Created a snapshot named WK1-PM1-Clean-Lab-Setup.
5. Added a description of the lab configuration.
6. Verified that the snapshot appeared in the snapshot list.

The snapshot provides a restore point for the configured virtual machine before future cybersecurity exercises.

**Screenshot 13:** *Insert a screenshot showing the saved Kali Linux snapshot in VirtualBox.*

## 11. FINAL VERIFICATION CHECKLIST

| Requirement | Status |
|---|---|
| **VirtualBox installed** | [Completed / Pending] |
| **Kali Linux imported and booted** | [Completed / Pending] |
| **Custom NAT Network created** | [Completed / Pending] |
| **Network subnet configured as 10.0.0.0/24** | [Completed / Pending] |
| **Kali Linux static IP configured as 10.0.0.2/24** | [Completed / Pending] |
| **Internet connectivity verified** | [Completed / Pending] |
| **Shared clipboard enabled and tested** | [Completed / Pending] |
| **Drag-and-drop enabled and tested** | [Completed / Pending] |
| **Windows Downloads folder shared and accessible** | [Completed / Pending] |
| **Virtual machine snapshot created** | [Completed / Pending] |

## 12. CONCLUSION

The WK1-PM1 task focuses on creating a virtualized cybersecurity testing environment using Oracle VirtualBox and Kali Linux.

The required lab configuration includes a custom NAT Network using the 10.0.0.0/24 subnet, the specified static IP address for Kali Linux, internet connectivity, host–guest integration, a shared Downloads folder, and a virtual machine snapshot.

The final status of the lab is based on the verification results and screenshots included in this report. Any requirement that remains incomplete will be identified and resolved before final submission.

**END OF REPORT**

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